GRA-Flow: Gain-Risk Aligned Rectified-Flow Refinement for Task-Free OOD Image Restoration

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ABSTRACT

Task-free image restoration requires deciding whether an additional correction will improve a pretrained restorer without observing a degradation label or a clean target. We study this decision for a frozen rectified-flow proposal and introduce GRA-Flow, Gain-Risk Aligned Flow. The change in image-space squared error is exactly quadratic in the continuous endpoint action, separating observable correction energy from unknown signed alignment with the baseline error. We derive an analytic action that is non-harmful whenever a predicted lower score satisfies its stated pointwise condition. A target-free trajectory predictor learns normalized directional benefit through location, sign, paired-intervention, and lower-tail supervision; a shared residual offset controls the resulting correction decision. Matched development experiments over 400 clean identities and five degradation environments separate prediction errors from offset-induced rejection. Saved-checkpoint replay identifies larger excluded-identity residual tails, particularly under low-light degradation. These results support a diagnosis of the current benefit predictor, not a claim of independently certified OOD improvement. The frozen fresh-shift panel is reported separately from independent risk certification and real-degradation evaluation.

1. Introduction

An additional restoration step should be judged against the image it changes. A rectified-flow proposal may improve a frozen restoration output on one degradation and damage it on another, even when both inputs are processed by the same model. Task-free out-of-distribution (OOD) restoration therefore requires more than producing a correction: it requires deciding whether that correction is beneficial and how strongly to apply it without a degradation label or target-domain tuning. Flow matching and rectified flow provide a vector-field construction for image transport [1, 2]. We build on a fixed restoration proposal and study a different decision: continuous interpolation between the frozen baseline and the proposal endpoint. The unknown quantity is the alignment between the proposed correction and the baseline's remaining error. Its sign determines whether any positive correction can improve squared error; its magnitude determines the optimal action within the permitted interval. This leads to our research question: *How can we improve restoration quality while controlling the risk of degrading the baseline output?* We develop GRA-Flow, Gain-Risk Aligned Flow, to make this decision from target-free trajectory observables. The method separates three components: an exact image-space response model, a predictor of normalized directional benefit, and a shared one-sided offset followed by an analytic action. We use paired degradation interventions on the same clean identity to supervise changes in benefit while keeping identity groups disjoint during evaluation. This is an intervention-informed prediction design, not identification of a latent causal graph. Our analysis establishes the exact quadratic correction response and a deterministic sufficient condition for non-harmful action. The empirical question is whether observable trajectory information estimates the required directional benefit well enough to retain useful correction coverage under a shared rule. The current development evidence separates prediction error from offset-induced rejection through matched endpoint-only and objective controls, environment-level residual diagnostics, and training-versus-OOF replay. The frozen fresh-shift panel is reported separately from independent risk certification and real-degradation evaluation; development results are not presented as their substitute. The study has three components: an explicit baseline-relative decision target; an implemented trajectory predictor with paired-intervention supervision; and a diagnostic protocol that reports restoration quality together with correction coverage, magnitude, and harmful counts. The theoretical component is exact under its stated image-space assumptions, whereas representation superiority and statistical risk control are evaluated as empirical claims rather than assumed properties of the architecture.

2. Related Work

All-in-one restoration. Shared restoration models use prompts, adaptive processing, routing, and generative representations to address multiple degradations [3–11]. We keep the restoration backbone fixed and evaluate the subsequent correction decision relative to its output.

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Historical comparisons with these methods remain restoration context, not evidence that the current selector dominates every model.

Flow-based refinement and guidance. Flow matching regresses vector fields of conditional paths, and rectified flow uses straight interpolation paths [1, 2]. Flow-based image restoration includes plug-and-play and degradation-flow approaches [12, 13]. Classifier-free guidance controls the vector field using conditional and unconditional predictions [14]; manifold-projection guidance has also been studied for flow matching [15]. Our endpoint action is different: after a fixed trajectory has been computed, it selects a point on the segment between its baseline and final proposal. A trajectory guidance scale and this endpoint action are not interchangeable.

Multi-environment learning. Environment-based learning includes objectives such as invariant risk minimization [16]. We use known degradation interventions to construct matched observations of one clean identity and supervise differences in correction benefit. Environment labels organize training and residual diagnostics but are not inputs to the deployed predictor. This construction supports controlled comparisons within the specified intervention family; it does not establish causal sufficiency or a distribution-free guarantee on arbitrary unseen degradations.

Selective prediction and calibration. Selective prediction evaluates risk together with coverage, and SelectiveNet jointly learns prediction and rejection [17]. Conformalized quantile regression separates quantile estimation from calibration of prediction intervals [18]. Conformal risk control addresses expected monotone losses, while Learn then Test formulates risk calibration through hypothesis testing [19, 20]. RACER applies risk-calibrated set selection to LLM routing [21]. Our decision object is a continuous image correction relative to a frozen baseline. Its deterministic error response motivates a conservative score, whose statistical reliability must be assessed separately from prediction accuracy.

3. Problem Formulation

Let X be a degraded image, Y its clean target, and $Y_0 = F_0(X)$ the output of a frozen restorer. A fixed rectified-flow sampler produces a proposal endpoint Y_f and trajectory observables $\Phi(X)$. We hold both the restorer and the proposal model fixed while learning an action $\alpha(X) \in [0, 1]$. The corrected output is

$$Y_\alpha = Y_0 + \alpha(Y_f - Y_0). \tag{1}$$

Thus $\alpha = 0$ retains the baseline, whereas $\alpha = 1$ applies the complete proposal. The action scales the completed image-space correction; it does not change the trajectory guidance scale.

We measure correction utility relative to the same baseline using

$$\Delta\mathrm{MSE}(\alpha) = \mathrm{MSE}(Y_\alpha, Y) - \mathrm{MSE}(Y_0, Y). \tag{2}$$

A negative value indicates improvement, and a positive value defines a harmful correction. For a specified evaluation distribution and action rule, we distinguish marginal harmful risk, correction coverage, and harmful risk among accepted corrections:

$$\begin{aligned} R_{\mathrm{marg}} &= \Pr\{\Delta\mathrm{MSE}(\alpha) > 0\}, \\ C &= \Pr\{\alpha > 0\}, \\ R_{\mathrm{acc}} &= \Pr\{\Delta\mathrm{MSE}(\alpha) > 0 \mid \alpha > 0\}. \end{aligned} \tag{3}$$

For $C > 0$, $R_{\mathrm{marg}} = C R_{\mathrm{acc}}$; at $C = 0$, R_{acc} is undefined. These quantities distinguish useful selective correction from rejecting every proposal. We seek restoration gains together with useful coverage and low harmful risk, reporting the marginal and accepted risks separately. This distinction parallels the risk–coverage formulation of selective prediction [17], with rejection here returning an existing restoration output.

Clean targets are available only for offline supervision and evaluation. At inference, the action uses the degraded input, frozen model outputs, and fitted prediction statistics, without a clean target or degradation label. The following section identifies the unknown quantity that determines correction utility and constructs a target-free rule for estimating and applying it.

4. Gain-Risk Aligned Flow

In this section, we present Gain-Risk Aligned Flow (GRA-Flow) for baseline-relative correction with a frozen proposal. GRA-Flow estimates directional benefit from target-free trajectory observables and uses a shared residual offset to determine whether and how strongly to apply the correction. We first derive the decision target, then describe its prediction and residual adjustment, and finally specify the inference rule.

4.1. From Correction Error to Directional Benefit

The first task is to determine what makes a proposed correction beneficial. For the pixel-averaged inner product $\langle v, z\rangle_p = D^{-1}\sum_{j=1}^{D} v_j z_j$, define

$$\begin{aligned} d &= Y_f - Y_0, \quad r = Y - Y_0, \\ a &= \langle d, d\rangle_p, \quad b = \langle d, r\rangle_p. \end{aligned} \tag{4}$$

The correction energy a is observable, but its signed alignment b with the remaining baseline error depends on the unknown target. For $a > 0$, write $c = b/\sqrt{a}$ for normalized directional benefit and $u = \alpha\sqrt{a}$ for the applied correction magnitude.

Proposition 1 (Exact correction response)**.** *For fixed Y_0, Y_f, Y, the change in pixel-averaged squared error is*

$$\Delta\mathrm{MSE}(\alpha) = a\alpha^2 - 2b\alpha = u^2 - 2cu. \tag{5}$$

For $a > 0$, its constrained minimizer is $\alpha^\star = \mathrm{clip}(b/a, 0, 1)$. If $b \le 0$, no positive action decreases squared error. If $b > 0$, every $0 < \alpha < \min(1, 2b/a)$ decreases squared error; the endpoint $\alpha = 1$ also improves it when $a < 2b$.

This response separates correction magnitude from correction usefulness. The sign of c determines whether a beneficial positive action exists; its magnitude determines how far to move along the proposal. The normalized benefit c is a signed projection of the baseline error, rather than the realized error reduction, which also depends on u. The oracle action $\alpha^\star$ uses the clean target and serves only as an offline diagnostic. Zero-energy proposals have $d = 0$ and receive $\alpha = 0$.

Equation (5) reduces the decision to estimating one unknown scalar from observable information. We therefore learn c from the frozen trajectory and account for overestimation before converting the prediction into an action.

4.2. Learning Directional Benefit from Paired Interventions

We estimate directional benefit from $\Phi(X)$, a summary of the same frozen proposal pass that produces Y_f. The observables describe state and correction magnitudes, relations between the reference and increment velocity fields, and changes between consecutive steps. They also include input–baseline and endpoint-correction magnitudes. All features are available without the clean target or an environment label. The complete feature definition is given in Section 9.1.

For each clean identity i, known degradation interventions generate $X_{i,e}$ and offline labels $c_{i,e}$. All derivatives of an identity remain in the same fold. Feature standardization and target location/scale are fitted only on the training portion of each fold. With $\widetilde{c} = (c - m_c)/s_c$, the predictor produces a location output m_θ, an auxiliary sign logit s_θ, and a raw gap g_θ. Its lower-tail output is parameterized as

$$\widetilde{q}_{\theta,\tau} = m_\theta - \mathrm{softplus}(g_\theta). \tag{6}$$

The location output estimates the central tendency of directional benefit, while the gap permits a lower prediction. We train the lower-tail output with the pinball loss used in quantile regression [18]. Predictions are returned to the original benefit units by $q_{\theta,\tau} = m_c + s_c \widetilde{q}_{\theta,\tau}$; we abbreviate this output as q_θ below.

Paired supervision targets how benefit changes when the degradation changes while clean identity is held fixed. Let h be the unit-threshold Huber loss. We compare all unordered intervention pairs within each training identity:

$$\mathcal{L}_{\mathrm{pair}} = \operatorname*{mean}_{i,e<e'} h\big[(m_{i,e} - m_{i,e'}) - (\widetilde{c}_{i,e} - \widetilde{c}_{i,e'})\big]. \tag{7}$$

This objective supervises differences in directional benefit rather than enforcing identical representations across degradations. It uses controlled intervention pairs as training data; the deployed predictor does not require the intervention identity.

The complete training objective combines location, sign, paired-difference, and lower-tail supervision:

$$\mathcal{L} = \mathcal{L}_{\mathrm{loc}} + 0.5\mathcal{L}_{\mathrm{sign}} + 0.5\mathcal{L}_{\mathrm{pair}} + \mathcal{L}_{\mathrm{tail}}. \tag{8}$$

Here $\mathcal{L}_{\mathrm{loc}}$ applies h to $m_\theta - \widetilde{c}$, and $\mathcal{L}_{\mathrm{sign}}$ uses binary cross-entropy with label $\mathbf{1}\{c > 0\}$. The lower-tail term applies $\rho_\tau(z) = \max(\tau z, (\tau - 1)z)$ to $z = \widetilde{c} - \widetilde{q}_{\theta,\tau}$. Each of these three terms averages first within an environment and then equally across environments. The sign branch supplies auxiliary supervision and does not form an additional inference gate.

Huber supervision makes m_θ a robust location estimate, not necessarily the conditional mean. Likewise, the lower-tail parameterization and pinball loss do not establish a pointwise lower bound on c. The learned score may still overestimate benefit, motivating the residual adjustment below.

4.3. Constructing a Conservative Benefit Score

The next task is to account for one-sided prediction error before choosing a correction magnitude. A positive residual

$$R_j = q_j - c_j \tag{9}$$

indicates that directional benefit was overestimated. We therefore use the upper tail of these residuals to adjust the lower-tail predictor. Separating prediction from residual adjustment follows the principle of conformalized quantile regression [18]; the statistical validity of an adjustment additionally depends on how its data and scores are constructed.

In the reported development procedure, q_j is the held-fold prediction in original benefit units. Within environment e, let $R_{e,(k_e)}$ denote the k_eth ordered residual, where $k_e = \lceil (n_e + 1)0.95 \rceil$. If $k_e > n_e$, its value is defined as $+\infty$. We construct

$$\delta = \max_e R_{e,(k_e)}, \qquad \ell(X) = q_\theta(\Phi(X)) - \delta. \tag{10}$$

Taking the maximum uses the most conservative environment-specific residual offset among the observed environments. The resulting δ is shared across inputs, so applying ℓ requires no environment-specific threshold at inference. The prediction and offset use the same unstandardized c units. The training quantile τ, residual order-statistic level, and action α have distinct roles.

The current offset reuses OOF development outcomes for diagnosis and candidate assessment. It is an empirical residual adjustment, not independent split-conformal certification. Conformal risk control and Learn then Test provide statistical guarantees under their respective loss and calibration assumptions [19, 20]; those guarantees do not follow from Eq. (10) alone. In particular, taking a maximum over observed environments does not certify arbitrary unseen degradations. Model validation, risk calibration, untouched certification, and fresh testing retain separate roles in Section 5.1.

4.4. Continuous Correction and Inference

Given the adjusted score, we choose a magnitude within the available correction segment. For $a > 0$, define

$$u = \mathrm{clip}(\ell, 0, \sqrt{a}), \qquad \alpha = u/\sqrt{a}. \tag{11}$$

A non-positive score retains the baseline. A score between zero and $\sqrt{a}$ applies a partial correction, and a score at least $\sqrt{a}$ applies the complete proposal. For $a = 0$, we set $\alpha = 0$ without dividing by a numerical floor. The final output follows Eq. (1).

Proposition 2 (Sufficient condition for non-harmful correction)**.** *For $a > 0$, if $\ell \le c$, the action in Eq. (11) satisfies $\Delta\mathrm{MSE}(\alpha) \le -u^2 \le 0$.*

The condition links score conservatism to image-space behavior. Its violation determines how prediction error can translate into harmful correction.

Proposition 3 (Sensitivity to lower-score violation)**.** *For the action in Eq. (11), define the one-sided score error $\eta = (\ell - c)_+$. Then*

$$\Delta\mathrm{MSE}(\alpha) \le -u^2 + 2u\eta. \tag{12}$$

Consequently, an accepted correction can be harmful only if $\eta > u/2$.

Thus ranking beneficial directions is insufficient: harmful correction also depends on score overestimation relative to the applied magnitude. These statements concern squared error on the specified image segment. They do not assert that the learned score satisfies $\ell \le c$ on every image. If a separate valid argument established $\Pr(\ell > c) \le \varepsilon$ on a specified distribution, it would imply $R_{\mathrm{marg}} \le \varepsilon$. For coverage $C > 0$, the corresponding accepted-risk implication is only

$$R_{\mathrm{acc}} \le \min(1, \varepsilon/C). \tag{13}$$

The proofs are given in Section 8.

Algorithm 1 summarizes inference for a fixed predictor and offset. The restorer, proposal model, predictor, standardization statistics, and shared offset are fixed before applying the rule. Inference uses neither a clean target nor a degradation label. Trajectory summaries are collected during the existing proposal pass, so the selector adds computation without requiring a second trajectory solve.

Algorithm 1 GRA-Flow correction with a fixed predictor and offset

Input: Degraded image X; frozen restorer F_0 and proposal sampler; fitted predictor q_θ with its standardization statistics; shared offset δ
Output: Restored image Y_{out} and action α

$Y_0 \leftarrow F_0(X)$
Compute Y_f and $\Phi(X)$ in one proposal pass
$d \leftarrow Y_f - Y_0$, $a \leftarrow \langle d, d \rangle_p$
if $a = 0$ **then**
    Return $(Y_0, 0)$
end if
Evaluate $q_\theta(\Phi(X))$ in original benefit units
$\ell \leftarrow q_\theta(\Phi(X)) - \delta$
$u \leftarrow \mathrm{clip}(\ell, 0, \sqrt{a})$
$\alpha \leftarrow u/\sqrt{a}$
$Y_{\mathrm{out}} \leftarrow Y_0 + \alpha d$
Return $(Y_{\mathrm{out}}, \alpha)$

5. Experiments

Evidence map. We separate the primary decision evidence from explanatory mechanism analyses. The primary evidence first quantifies the opportunity and failure modes of a fixed proposal, then asks whether a target-free score changes the gain–risk relation while the action amplitudes are held fixed. Mechanism analysis subsequently examines quantile sensitivity, predictor and paired-intervention controls, and output-space error decompositions. Frozen-policy transfer and historical stress tests remain separate from development diagnostics. Each comparison changes one source of variation while keeping the frozen restorer, proposal, identities, and metric fixed. We investigate when a fixed rectified-flow proposal should be applied to a frozen restoration output, and how strongly it should be applied without observing a task identifier or a target image. The evidence is organized around correction utility, benefit representation, and the risk–coverage trade-off. We distinguish the current Gain-Risk Aligned Flow endpoint policy from the earlier guidance-scale experiments retained in the supplementary results. The output action α is not the trajectory guidance scale w.

5.1. Evaluation Scope and Evidence Separation

The current development comparisons keep the restoration baseline and A+B proposal fixed. All transformations of one clean identity remain together during splitting. The registered corpus allocates 400 identities to scorer training, 150 to model validation, 150 to risk calibration, 300 to independent certification, and 200 to fresh testing. The OOF experiments reported below use only the 400 scorer-training identities, with Gaussian blur, Gaussian noise, JPEG, low-light gamma, and spatial haze interventions. They do not evaluate independent certification or fresh-shift performance. We report absolute PSNR alongside baseline-relative changes, averaging per-image scores within environments and then weighting environments equally. Correction coverage is the fraction with $\alpha > 0$; correction mass is the all-image mean of α. Harmful/accepted counts accompany conditional harmful fractions, and no-acceptance cases have an undefined harmful fraction rather than zero risk. An oracle using the clean target is admissible only as an offline diagnostic, not as a deployable comparator.

Predictor implementation and matched controls. We use a two-layer [128, 64] SiLU predictor with the objective in Eq. (8). The recorded training setting is 300 epochs of

full-batch AdamW, with learning rate 10^{-3} and weight decay 10^{-3}. The original controls use fixed $\tau \in \{0.05, 0.10\}$, with additional quantile settings reported in the sensitivity study. Endpoint-only retains the same input width and predictor capacity but masks all coordinates except input–baseline RMS, endpoint-correction RMS, and the final-step summaries. Single-loss controls zero one weight in Eq. (8). Mean-only keeps the location term and uses its location prediction directly in Eq. (10); it therefore differs from removing pinball supervision while retaining the gap head. The latter does not directly supervise the gap and is treated as a diagnostic control.

5.2. Primary evidence: proposal opportunity and target-free selection

We next isolate two possible explanations for unreliable correction: the inference setting and the usefulness of the proposed direction itself. We freeze the backbone and refiner checkpoints and evaluate four settings on the same 400 clean identities under five controlled degradations (2,000 records per setting). The current setting uses $w = 1.09785$ and 10 integration steps; the controls set $w = 0$, set $w = 1$, or double the integration steps to 20. The current endpoint is replay-checked against the saved records, and the backbone outputs are shared across settings. These are paired development diagnostics, not an additional independent OOD benchmark. Imaging and weather degradations form the main panel; JPEG remains a separate compression-transfer probe.

Guidance and integration do not explain all environment differences. Table 1 shows a clear benefit of refinement for noise and blur. For noise, the current proposal raises PSNR from 27.833 to 29.266 dB and SSIM from 0.7241 to 0.8163; removing conditional guidance gives 28.871 dB and 0.7895. Blur improves under all four settings. In contrast, doubling the integration steps does not recover the backbone quality in low-light or spatial haze, and replacing the current guidance scale with $w = 1$ produces similar outcomes. The $w = 0$ low-light control reaches 17.505 dB, close to the 17.499 dB backbone, while its SSIM is 0.7925 versus 0.7942. Thus, these controls localize a degradation-dependent correction issue that is not resolved by the tested guidance and step-count changes.

Distinguishing overshoot from an unhelpful direction. With d and r defined as in the endpoint analysis, we evaluate $\alpha^\star = \mathrm{clip}(b/a, 0, 1)$ for $a > 0$ and choose zero when $a = 0$. Here α scales the completed correction and is distinct from guidance w. Table 2 shows that blur has 393 positive directions and only 22 harmful full steps; the full proposal already captures most of its segment-wise oracle gain. Noise retains more shrinkage headroom: the oracle reaches 29.500 dB versus 29.266 dB for the full proposal. Low-light and haze have only 160 and 180 positive directions, respectively, while the corresponding oracle PSNRs are 17.536 and 29.033 dB, close to their backbone values of 17.499 and 28.955 dB. This distinction matters for our research question: a decision rule can suppress harmful actions or reduce overshoot, but cannot create a beneficial direction where none exists on the proposed segment. The evidence therefore motivates diagnosing the residual components of these fixed proposals alongside learning when to apply them. It does not by itself establish causal representation quality or independent risk control.

Table 1
Frozen inference controls on matched development identities. Absolute PSNR (dB) and SSIM are averaged over 400 images per environment. Current uses $w = 1.09785$. All refiner settings share the same checkpoint; only guidance or integration steps change. JPEG is an additional compression-transfer test. Rows group Gaussian blur, Gaussian noise, low-light, spatial haze, and JPEG transfer.

Degradation	w	Steps	BaryIR		GRA-Flow	
			PSNR	SSIM	PSNR	SSIM
Blur	1.098	10	25.42	0.741	25.69	0.763
	0	10			25.68	0.759
	1	10			25.69	0.762
	1.098	20			25.70	0.764
Noise	1.098	10	27.83	0.724	29.27	0.816
	0	10			28.87	0.789
	1	10			29.26	0.816
	1.098	20			29.22	0.814
Low-light	1.098	10	17.50	0.794	17.38	0.785
	0	10			17.50	0.792
	1	10			17.39	0.786
	1.098	20			17.36	0.784
Haze	1.098	10	28.95	0.946	28.36	0.939
	0	10			28.41	0.942
	1	10			28.38	0.940
	1.098	20			28.35	0.938
JPEG	1.098	10	26.91	0.815	26.53	0.804
	0	10			26.63	0.808
	1	10			26.54	0.805
	1.098	20			26.51	0.804

5.3. Directional Information and Correction Decisions

We distinguish three questions: whether trajectory observables rank beneficial directions, whether an action optimizer avoids harmful corrections, and whether a target-free direction score can filter a fixed action. All comparisons below use the same 400 development identities and five identity-grouped OOF folds; none uses independent certification or fresh-shift outcomes.

Trajectories improve pooled benefit-direction ranking. We evaluate the auxiliary sign score against $\mathbb{1}\{b > 0\}$ using average precision (reported as AUPRC). The matched full-trajectory and endpoint-only predictors retain the same 92-dimensional interface, hidden widths, objective, and training budget. At $\tau = 0.05$, pooled AUPRC is 0.8563 for the full trajectory and 0.7876 for the endpoint-only control; at $\tau = 0.10$, the corresponding values are 0.8558 and 0.7878.

Table 2
Direction and oracle-amplitude analysis for the current 10-step proposal. Absolute PSNR/SSIM are shown before and after correction. Positive direction counts $b > 0$; full-step harm counts $\Delta\text{MSE} > 10^{-12}$ at $\alpha = 1$ (400 images per environment). The ground-truth oracle minimizes MSE on the fixed segment $\alpha \in [0, 1]$: it is a diagnostic PSNR upper bound on this segment, not a deployable policy or an SSIM optimum. Degradation rows follow Table 1.

Degradation	BaryIR		Full		Oracle		Positive direction	Full-step harm
	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM		
Blur	25.42	0.741	25.69	0.763	25.70	0.762	393/400	22/400
Noise	27.83	0.724	29.27	0.816	29.50	0.822	324/400	115/400
Low-light	17.50	0.794	17.38	0.785	17.54	0.794	160/400	281/400
Haze	28.95	0.946	28.36	0.939	29.03	0.946	180/400	316/400
JPEG	26.91	0.815	26.53	0.804	26.92	0.814	67/400	349/400

The paired differences are +0.0687 and +0.0680, with 95% percentile intervals [0.0536, 0.0840] and [0.0530, 0.0836]. We resample 400 clean identities with all five associated environments together, using 2,000 bootstrap replicates of the fixed OOF predictions. These intervals describe sample-level variation conditional on the fitted predictions, not variation from retraining or independent confirmation.

Figure 1(a,b) reports every environment and its positive-class fraction. At $\tau = 0.05$, the JPEG AUPRC changes from 0.3381 to 0.4253, whereas low-light changes from 0.5939 to 0.5811. Thus, the pooled advantage is not an improvement in every environment. Moreover, the paired location-based action-regret intervals include zero at both settings. We therefore attribute the observed advantage to benefit-direction ranking, without equating it with a demonstrated advantage in calibrated correction quality.

Average action utility and directional reliability are distinct. To examine the decision objective, we fit a separate action-regret control on the same trajectory features and folds. Its sigmoid output specifies $\alpha \in (0, 1)$ and minimizes the exact excess response relative to $\alpha^\star$:

$$\mathcal{R}(\alpha) = a\alpha^2 - 2b\alpha - \left[a(\alpha^\star)^2 - 2b\alpha^\star\right]. \tag{14}$$

Training equally weights environments and divides the loss by the training-fold mean of a, a single positive normalization constant. This is a diagnostic objective replacement, not an additional component of the lower-score policy. Its strictly positive output explains its 100% coverage; that coverage is not evidence of successful selection.

The control attains a Macro PSNR gain of +0.3256 dB, with 853 harmful images. Of these, 816 have the wrong direction and 37 overshoot along a beneficial direction (Table 3). Holding its predicted amplitudes fixed and rejecting only cases with GT-derived $b \leq 0$ yields +0.3599 dB and 37 harmful cases among 1,124 accepted corrections. This target-informed control isolates the contribution of direction errors while preserving the remaining amplitude errors; it is not a deployable gate. It differs from the full GT action oracle, which selects $\alpha^\star$ and attains +0.4142 dB with no harmful accepted corrections.

A target-free sign score can filter fixed actions. We next replace the GT direction test with a predicted score. The auxiliary gate uses the existing direct-quantile predictor's sign head at $\tau = 0.02$. A dedicated control fits the same-width predictor with sign BCE alone. Both act on the same frozen action-regret amplitudes, accepting only when the sigmoid sign score exceeds a fixed cutoff $p_0 \in \{0.5, 0.9\}$. These scores are not assumed to be calibrated probabilities. The action predictor and sign predictor are separate models; this diagnostic adds a scorer and does not claim unchanged total inference cost.

Table 3 reports both cutoffs. At $p_0 = 0.9$, the auxiliary gate retains 449 corrections, including 32 harmful cases, while achieving +0.3322 dB over the frozen baseline. Relative to applying all learned actions, it filters many harmful corrections while retaining the all-image average gain. The dedicated gate accepts 588 cases with 68 harmful corrections and +0.3342 dB. Because these are different operating points, the larger accepted set alone does not establish a better risk–utility trade-off. Figure 1 presents the complete saved threshold sweeps rather than selecting only a favorable cutoff. Rejected images remain at the baseline and contribute to every reported PSNR average. The curves are descriptive OOF diagnostics, not a certified deployment rule.

Together, these controls separate information available in the trajectory, errors in the sign of correction benefit, and the residual cost of applying an excessive positive action. Additional parameterization, residual-scale, and spatial-observable diagnostics are reported in Section 10; they are not substituted for a frozen-policy OOD evaluation.

5.4. Mechanism analysis

The following analyses explain the conditions under which the primary decision problem is difficult; they are not independent performance claims. They test the separation implied by Proposition 3: ranking, lower-score conservatism, action magnitude, and output-space error attribution are distinct quantities.

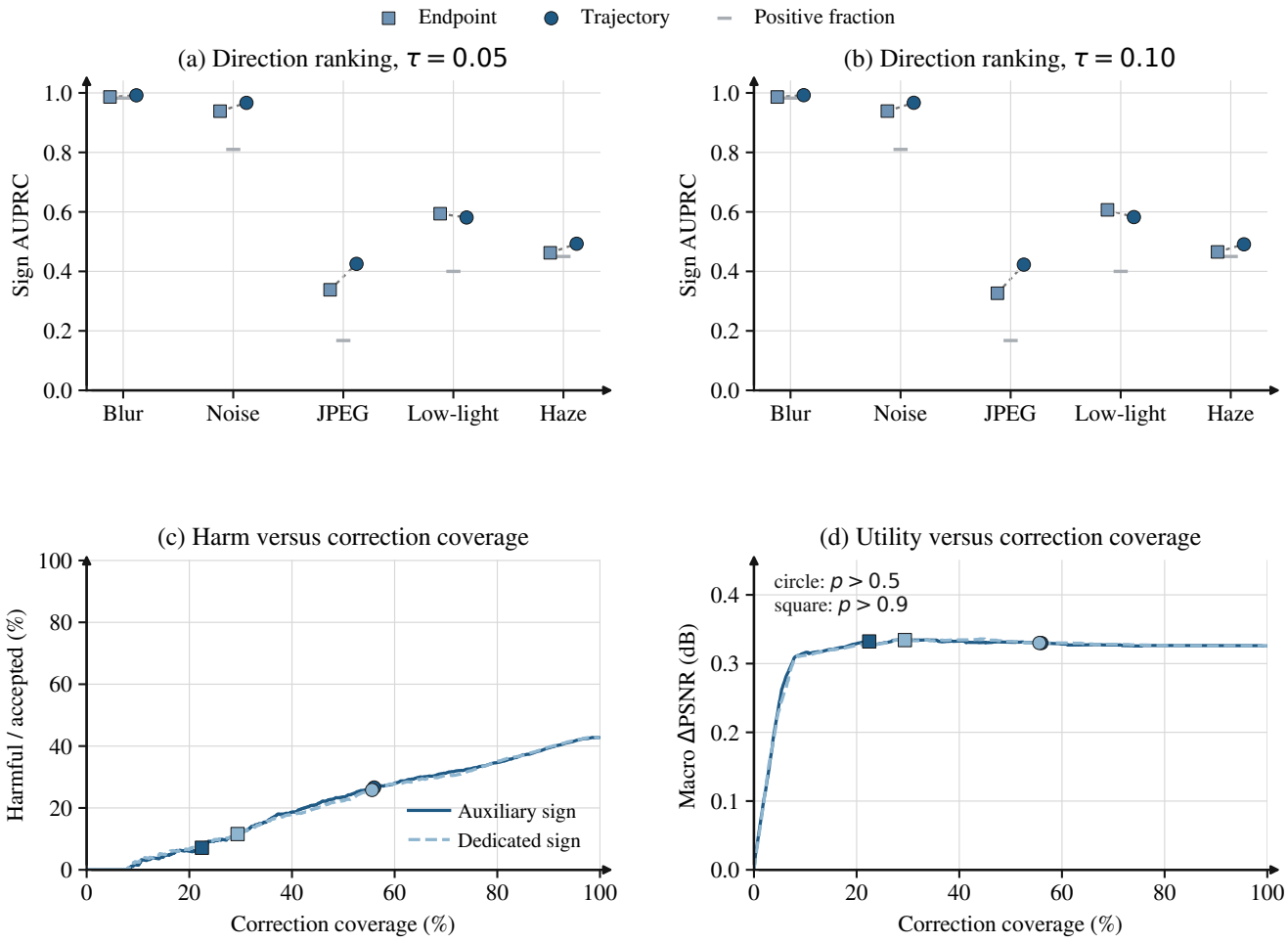


Fig. 1. Development-only mechanism diagnostics on 2,000 fixed OOF records. (a,b) Matched endpoint-only and full-trajectory predictors are compared by benefit-direction ranking at $\tau = 0.05$ and $\tau = 0.10$; paired connectors compare predictors within each environment, and gray ticks mark the positive-class fraction. (c,d) The same saved threshold sweep for auxiliary and dedicated sign gates reports the harmful fraction among accepted corrections and the baseline-relative Macro PSNR gain, respectively. Solid dark-blue and dashed light-blue curves denote auxiliary and dedicated gates; circles and squares indicate the predeclared $p_0 = 0.5$ and $p_0 = 0.9$ cutoffs. Every environment and saved threshold point is retained. These fixed-OOF diagnostics describe ranking and decision trade-offs, not fresh-shift confirmation or calibrated conditional risk.

Table 3
Fixed-action direction-gate diagnosis on 2,000 OOF development records. PSNR denotes Macro PSNR (dB); parenthesized values are paired changes relative to the frozen baseline before rounding, with upward arrows indicating increases. Coverage is the percentage of all records accepted; mean action is the mean of α over all records. Harmful / accepted reports raw harmful/accepted counts, which do not constitute a population-level guarantee. Both fixed score cutoffs are reported. The GT-direction filter is an offline diagnostic that retains the learned amplitudes.

Method	Setting	Coverage (%)	Mean action	Harmful / accepted	PSNR (Δ)
Frozen baseline	–	0.00	0.0000	–	25.32 (0.00)
Action-regret control	Ungated	100.00	0.3536	853/2000	25.65 (0.33↑)
Auxiliary sign	$p_0 = 0.5$	56.00	0.3207	298/1120	25.65 (0.33↑)
	$p_0 = 0.9$	22.45	0.1990	32/449	25.66 (0.33↑)
Dedicated sign	$p_0 = 0.5$	55.60	0.3038	287/1112	25.65 (0.33↑)
	$p_0 = 0.9$	29.40	0.2292	68/588	25.66 (0.33↑)
GT-direction filter	Diagnostic	56.20	0.2874	37/1124	25.68 (0.36↑)
GT action oracle	Diagnostic	56.20	0.4575	0/1124	25.74 (0.41↑)

5.4.1. Quantile sensitivity of the lower-score rule

We separate four empirical questions: what the current endpoint policy achieves, why a fixed proposal needs selective application, which prediction information supports that decision, and which correction components explain the response. The current method is the lower-tail C predictor with the shared residual offset and analytic amplitude defined in the method section. Its matched controls appear in Table 4. The action-regret predictors, sign gates, and sign-coupled pairing controls are separately identified development models. We use them to test specific explanations, not as alternative rows of a single final method. Target-informed amplitude controls measure available correction capacity; they do not estimate deployable performance. Earlier restoration models and their qualitative comparisons remain in the historical supplementary section. This separation prevents combining favorable measurements from different models into one Ours result.

5.4.2. Trajectory predictors and paired-intervention controls

We test whether the low correction coverage in Section 9.3 can be explained by a single auxiliary loss or by the use of trajectory features. We retain the same 400 clean identities, five grouped folds, frozen proposal, $[128, 64]$ predictor, and 300-epoch AdamW budget. Each single-loss ablation sets exactly one objective weight to zero. The endpoint-only control masks non-endpoint feature coordinates while preserving the 92-dimensional input interface and predictor capacity. Table 4 reports the full, without-pairwise, and endpoint-only predictors at four quantile levels, $\tau \in \{0.02, 0.05, 0.10, 0.20\}$, using the same shared-offset construction and analytic action. The complete auxiliary-loss controls at the original two levels are retained in Table 10; none of these comparisons selects a deployment policy.

Low coverage persists across auxiliary-loss controls. Tables 4 and 10 shows that removing pairwise supervision changes coverage from 4.35% to 4.85% at $\tau = 0.05$, but from 3.85% to 3.70% at $\tau = 0.10$. Removing sign supervision likewise leaves coverage near the full predictor. Neither intervention reaches the fixed 50% coverage and 0.25 correction-mass requirements. We therefore find no evidence that deleting either auxiliary term alone resolves the shared-offset bottleneck. The endpoint-only control attains higher point estimates of PSNR gain at both settings, with no observed harmful accepted corrections, but coverage remains 4.80% and 3.85%. This comparison does not establish an advantage for the full trajectory representation under the present normalized-C objective; zero observed harm at such coverage is not a risk certificate.

Quantile supervision and the next diagnostic question. In Table 10, removing pinball supervision leads to rejection of every proposal at both settings. Because this intervention removes the loss that directly trains the lower-quantile output, we interpret it as a diagnostic of that output's training dependence, rather than independent evidence of mechanism superiority. Together with the environment-residual analysis, the matched controls narrow the next question to whether directional-benefit errors arise during fitting or on unseen clean identities. Appendix 9 examines training-fold and excluded-fold errors while retaining the proposal, feature schema, data partitions, and deployment rule.

The mean-only predictor changes several supervision terms jointly; its environment-level results are retained in Appendix 9, together with the shared-offset and fitting diagnostics.

5.4.3. Pairing controls at matched execution and action magnitude

To test whether a gain comes merely from applying more corrections, we reuse the completed pairing-control predictions and fix both the execution count and action magnitude. Each scorer ranks the same 2,000 saved development proposals by its mean normalized-benefit score. We apply the full proposal ($\alpha = 1$) to the top 1,000 records and leave all other outputs unchanged. Thus, every row has 50% execution coverage and correction mass 0.5; an independent auxiliary sign head does not determine the ranking. This is a retrospective budget-matched diagnostic, not a calibrated deployment threshold.

Table 5 separates score quality from execution volume. Sign-coupled same-identity prediction raises absolute PSNR from 25.48 to 25.52 dB while applying exactly the same number of full corrections; its gain over the original paired control is therefore not explained by greater coverage or larger action magnitude. However, cross-identity pairing reaches 25.53 dB with 553 harmful actions, compared with 25.52 dB and 551 for same-identity pairing. The comparison does not isolate a unique benefit of preserving clean identity in the pairing loss. Moreover, more than half of the executed full actions are harmful in every row: a positive average PSNR gain is not sufficient evidence of useful risk control. These controls test the pairing explanation without changing the current endpoint policy or opening an independent evaluation partition.

5.4.4. Image-space decomposition of correction error

We examine which part of the frozen correction accounts for its error change in low-light and spatial haze. For each RGB channel, we project the correction d and baseline error $r = y - x_0$ onto their spatial means, writing $d = \bar{d} + d_\perp$ and $r = \bar{r} + r_\perp$. The mean and centered components are orthogonal under the pixel-averaged inner product. Consequently, both the pre-/post-correction MSE and the response $\Delta\mathrm{MSE} = a - 2b$ split exactly into two additive terms. This is an output-space diagnosis: the centered component is not assumed to isolate semantic structure or a causal factor.

A small correction component can dominate the net error change. Table 6 separates correction magnitude from correction usefulness. In low-light, the channel-mean component contributes 0.502364×10^{-3} of the total 0.694132×10^{-3} MSE increase (72.4%); in spatial haze, it contributes

Table 4
Sensitivity to the predicted quantile level τ. Matched coupled-C predictors on 400 clean identities under five interventions (2,000 OOF records). Full uses all trajectory features; Endpoint retains endpoint features only. PSNR (dB) and SSIM are equal-environment all-image means; parenthesized values are paired changes relative to the frozen baseline before rounding, with upward arrows indicating increases. Coverage is the fraction of images with a nonzero correction, mean action is the all-image mean of α, and harmful/accepted reports raw counts. τ is a prediction quantile, not a target risk level. All settings use the same shared-offset rule; these are development diagnostics.

τ	Method	Coverage (%)	Mean action (α)	Harmful / accepted	PSNR (Δ)	SSIM (Δ)
–	Frozen baseline	0.00	0.0000	–	25.32 (0.00)	0.804 (0.000)
0.02	Full	5.85	0.0238	3/117	25.45 (0.12 ↑)	0.810 (0.006 ↑)
	w/o paired-intervention loss	6.15	0.0235	5/123	25.43 (0.11 ↑)	0.809 (0.005 ↑)
	Endpoint	5.30	0.0357	0/106	25.52 (0.20 ↑)	0.815 (0.011 ↑)
0.05	Full	4.35	0.0152	3/87	25.40 (0.08 ↑)	0.808 (0.003 ↑)
	w/o paired-intervention loss	4.85	0.0194	4/97	25.42 (0.09 ↑)	0.808 (0.004 ↑)
	Endpoint	4.80	0.0223	0/96	25.45 (0.13 ↑)	0.810 (0.006 ↑)
0.10	Full	3.85	0.0115	4/77	25.38 (0.05 ↑)	0.806 (0.002 ↑)
	w/o paired-intervention loss	3.70	0.0110	3/74	25.37 (0.05 ↑)	0.806 (0.002 ↑)
	Endpoint	3.85	0.0120	0/77	25.39 (0.07 ↑)	0.807 (0.003 ↑)
0.20	Full	3.30	0.0058	3/66	25.35 (0.02 ↑)	0.805 (0.001 ↑)
	w/o paired-intervention loss	2.85	0.0035	2/57	25.34 (0.01 ↑)	0.805 (0.001 ↑)
	Endpoint	3.25	0.0059	0/65	25.36 (0.03 ↑)	0.805 (0.001 ↑)

Table 5
Pairing controls with identical execution count and full-action magnitude. PSNR is the all-image, equal-environment mean (dB); ΔPSNR is relative to the same baseline (25.32 dB), computed before rounding. Harmful / executed gives harmful and executed counts. All four rows use saved predictions from the sign-coupled comparison, including its original paired control.

Control	PSNR (dB)	ΔPSNR	Harmful / executed
Original paired	25.48	+0.1531	562/1000
Sign-coupled, same identity	25.52	+0.1990	551/1000
Sign-coupled, no pairing	25.50	+0.1719	558/1000
Sign-coupled, cross identity	25.53	+0.2034	553/1000

0.178544×10^{-3} of 0.259309×10^{-3} (68.9%). These are ratios of aggregate changes, not averages of per-image ratios or fractions of harmed images. Despite its smaller energy than the centered component, the mean component has negative average alignment with the baseline error in both environments. Haze additionally illustrates overshoot: the centered component has positive average alignment, but its correction energy exceeds twice that alignment, producing a positive net error change. Thus, correction energy alone cannot determine whether refinement is beneficial; its alignment with the remaining error is essential, as predicted by the endpoint response.

Figure 2 provides a paired visual reading of the decomposition rather than replacing the full-cohort table. The spatially constant mean component is visually subtle, whereas the centered maps emphasize local changes. In both illustrated examples, the mean component reduces MSE enough to offset the increase from the centered component, yielding a net PSNR gain. SSIM decreases slightly in both cases. These sample-level mean-component benefits contrast with the average MSE increase across the 400 records in each environment. This sample-level variation reinforces the need to estimate correction benefit per input, rather than prescribing a fixed component removal from an aggregate result. All 800 endpoints reproduce the saved outputs and metrics exactly; the largest discrepancy in the additive decomposition is below 3.2×10^{-17}.

5.5. Spatial Diagnostics of the Frozen Proposal

We complement the cohort-level decomposition with spatial diagnostics of low-light enhancement and dehazing. The low-light identity is `00153235c7a8d88223b1`; the haze pair is `0001_0.8_0.2/0001`. All stages use aligned 256×256 crops. Both examples use the unchanged backbone and ten-step proposal with $w = 1.09785$. We examine the full endpoint Y_f ($\alpha = 1$), before the learned correction decision; these two distinct identities provide descriptive examples rather than a paired cross-degradation comparison.

Table 6
Exact error decomposition on all 800 development records: 400 matched clean identities under each of two degradations. All numerical columns use units of 10^{-3}; entries average per-image quantities. Before and After are the baseline and full-proposal MSE of each component. Correction energy a and signed alignment $2b$ explain the change $\Delta = a - 2b$; positive Δ denotes increased error. Total equals Channel mean plus Centered, before rounding.

Low-light			
Quantity	Total	Channel mean	Centered
Before MSE	26.995354	20.630318	6.365036
After MSE	27.689486	21.132682	6.556804
Energy a	0.201599	0.026147	0.175452
Alignment $2b$	-0.492533	-0.476216	-0.016316
ΔMSE	+0.694132	+0.502364	+0.191768
Spatial haze			
Quantity	Total	Channel mean	Centered
Before MSE	5.447795	4.037642	1.410153
After MSE	5.707104	4.216186	1.490918
Energy a	0.106288	0.016350	0.089939
Alignment $2b$	-0.153021	-0.162194	+0.009173
ΔMSE	+0.259309	+0.178544	+0.080765

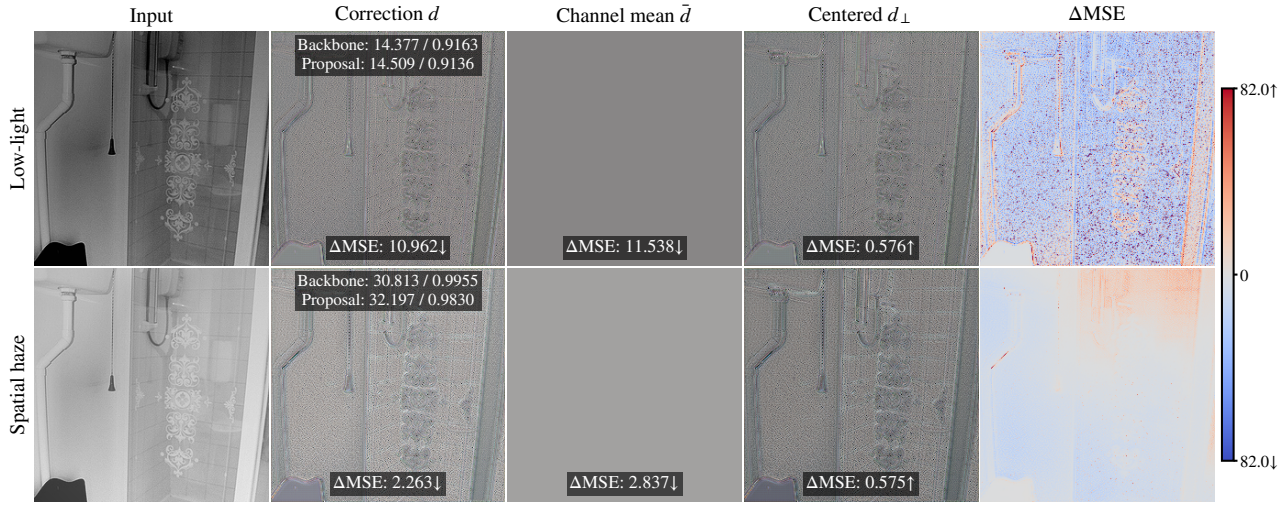


Fig. 2. Image-space correction decomposition. The same clean identity is shown under low-light and spatial haze, using aligned 256×256 crops. Columns show the degraded input, full correction d, channel mean $\bar{d}$, centered component $d_\perp$, and pixelwise MSE change (proposal minus backbone). Insets in the full-correction panels report backbone and proposal PSNR (dB)/SSIM. Component annotations and colorbar values are scaled by 10^4: ↑ denotes increased MSE and ↓ decreased MSE; red and blue indicate local increases and decreases, respectively. Component values are image-averaged contributions and sum to the full change before rounding. Correction RGB maps use $\mathrm{clip}(0.5 + c/(2L), 0, 1)$ with shared $L = 0.05$. The error map uses one symmetric range from the pooled 99.5th percentile of absolute pixelwise changes, with tails clipped for display. Metrics use unmodified tensors; no per-image contrast adjustment is applied.

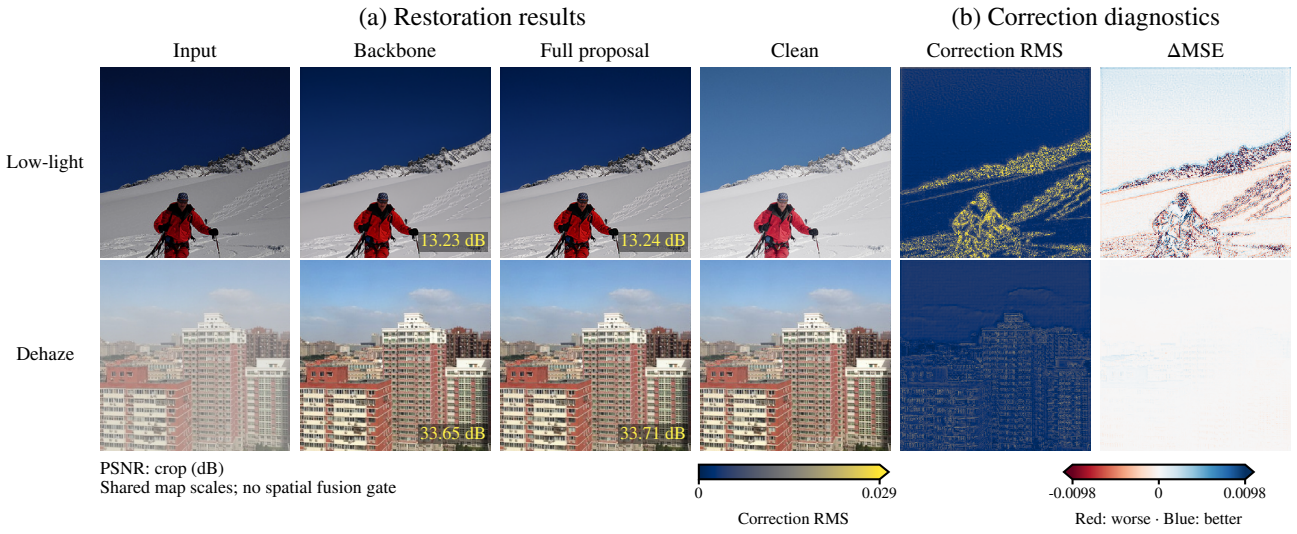


Fig. 3. Spatial correction and reconstruction error. (a) Low-light and haze inputs, frozen backbone outputs, full proposals and clean targets. Numbers report crop PSNR from the original floating-point outputs. (b) Correction RMS and local error reduction $R(p)$. The panel label ΔMSE uses the reduction convention: blue indicates lower error and red higher error, opposite to the error-increase convention of Eq. (5). Both rows share the pooled 99th-percentile RMS range and a symmetric pooled 99th-percentile absolute-R range; colorbar extensions indicate clipped tails. RGB clipping is for display only. Full proposal denotes $\alpha = 1$, not the selected policy output.

Correction magnitude and local benefit. Figure 3 separates the magnitude of the update from its effect on reconstruction error. For each pixel, we compute correction RMS as $\sqrt{\frac{1}{3}\sum_c d_c(p)^2}$ and error reduction as $R(p) = \frac{1}{3}\sum_c[(Y_{0,c}(p) - Y_c(p))^2 - (Y_{f,c}(p) - Y_c(p))^2]$. Thus, positive R denotes improvement, and its spatial mean equals $-\Delta\text{MSE}(1)$ in Eq. (5). The full proposal increases crop PSNR from 13.225 to 13.238 dB for low-light enhancement and from 33.648 to 33.710 dB for dehazing. Regions of increased and reduced error coexist within the crops, although the RGB endpoints look similar. This distinction makes local error maps useful for interpreting small aggregate changes without treating a large update as evidence of benefit.

Feature response under a common frozen encoder. We separately re-encode each input, backbone output and proposal with the same frozen BaryNet encoder, using its latent tensor after layer normalization ($384 \times 32 \times 32$). For each image, we fix the clean query at native coordinate $q = (16, 16)$ and compute $S_s(p) = \cos(F_s(p) - \mu, F_Y(q) - \mu)$ in the original 384-dimensional space. The shared mean μ is retained from balanced spatial-feature sampling on 32 development identities and is not refitted on these examples. Figure 4 shows that the input, backbone and proposal retain similar coarse query-similarity patterns in the low-light example, with visible changes near the sky–snow transition. Backbone and proposal responses are also close in the haze example. Together with Fig. 3, these observations show how modest endpoint improvements can accompany localized error changes while leaving the coarse feature response largely unchanged. These maps measure centered query similarity, not semantic labels, PCA colors or evidence of causal invariance; the clean query is used only for offline analysis.

5.5.1. *Component removal versus component-wise control*

We test whether the error decomposition in Section 5.4.4 supports a fixed component removal, or instead reveals a need to control component amplitudes separately. We retain all 400 development identities under both low-light and spatial haze, and construct six actions from the same saved baseline x_0 and correction $d = \bar{d} + d_\perp$. The four target-free controls apply no correction, the full correction, the channel mean alone, or the centered component alone. Two target-informed controls minimize per-image MSE: a shared amplitude applies $x_0 + \alpha d$, whereas separate amplitudes apply $x_0 + \alpha_{\text{m}}\bar{d} + \alpha_\perp d_\perp$, with every amplitude constrained to $[0, 1]$. Orthogonality makes the separate optimization two scalar quadratic problems; we solve them analytically, without a parameter sweep. Composed outputs are evaluated without clipping, preserving the exact quadratic response and the original metric implementation.

Aggregate error attribution does not prescribe a fixed removal. Table 7 directly tests this alternative explanation. Keeping only the channel mean gives 17.4313/0.7909 in low-light and 28.8878/0.9452 in haze, compared with backbone values of 17.4994/0.7942 and 28.9549/0.9459. Keeping only the centered component likewise does not improve the all-image means. Both components help some images and harm others: in haze, the mean-only correction improves 179 images but harms 221. Thus, a component's aggregate error contribution does not determine whether it should be removed for every input.

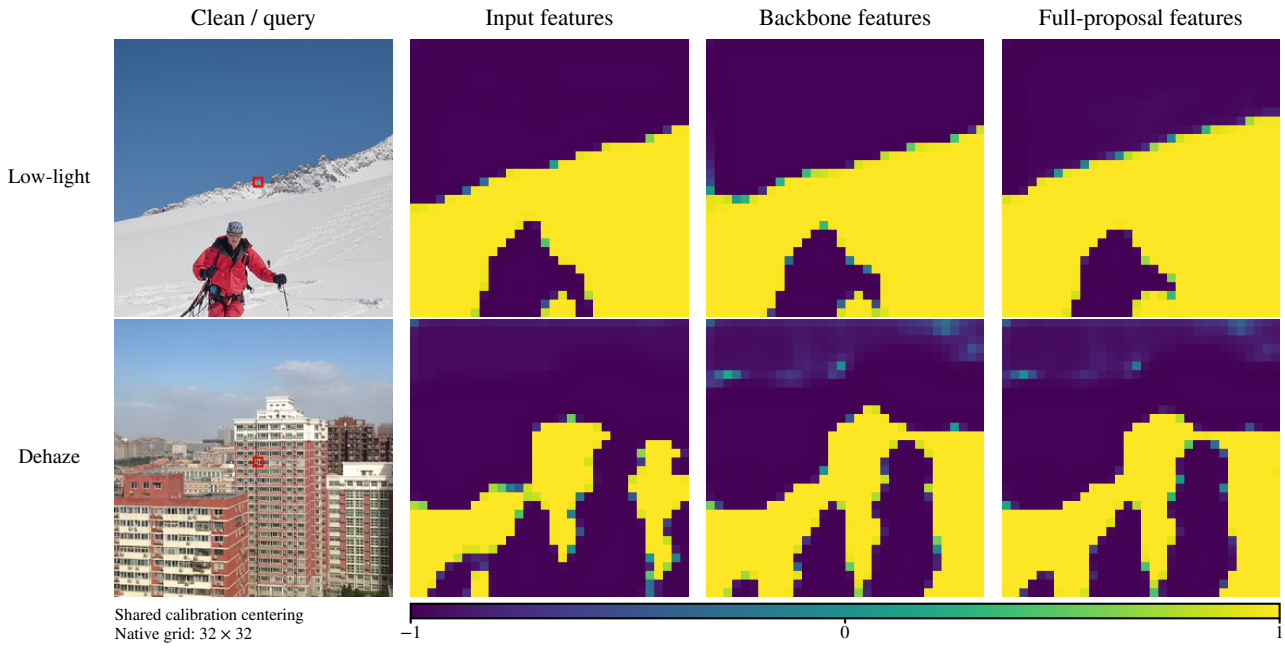


Fig. 4. Spatial feature response across degradation and restoration. All columns use the same frozen BaryNet encoder and latent-after-normalization layer, with 384 channels on a native 32×32 grid. The red square marks the fixed clean query cell. Colors show centered high-dimensional cosine similarity to that query, using one development-derived mean and a shared $[-1, 1]$ range. Nearest-neighbor enlargement preserves the native grid; no per-image contrast normalization is applied. The maps are an offline probe of the input and reconstructed images, not internal features of the 92-dimensional benefit predictor or a newly learned representation.

Table 7
Fixed component controls and target-informed amplitude bounds on matched development images. PSNR (dB) and SSIM are all-image means. Improved / harmed gives raw improved/harmed counts out of 400 images per environment, using a 10^{-12} MSE tolerance; remaining images are unchanged within this tolerance. The last two rows use clean targets to optimize MSE and are offline diagnostics, not a deployable method or an SSIM optimum.

Control	Low-light ($n = 400$)			Spatial haze ($n = 400$)		
	PSNR (dB)	SSIM	Improved / harmed	PSNR (dB)	SSIM	Improved / harmed
Backbone (no correction)	17.50	0.794	0/0	28.95	0.946	0/0
Full correction d	17.38	0.785	119/281	28.36	0.939	84/316
Channel mean only $\bar{d}$	17.43	0.791	134/266	28.89	0.945	179/221
Centered only $d_\perp$	17.44	0.787	99/301	28.43	0.939	65/335
Shared-amplitude oracle	17.54	0.794	160/0	29.03	0.946	180/0
Separate-amplitude oracle	17.55	0.795	241/0	29.13	0.947	274/0

Independent amplitudes expose additional usable correction directions. Separating the amplitudes raises haze PSNR from the shared oracle's 29.0326 to 29.1294 dB, while SSIM changes from 0.9457 to 0.9474. The number of improved images increases from 180 to 274. In low-light, PSNR changes from 17.5362 to 17.5534 dB and improved images increase from 160 to 241. Relative to the backbone, the separate oracle therefore achieves gains of 0.1745 dB in haze and 0.0540 dB in low-light. These results quantify correction capacity hidden by tying the two amplitudes; they do not require changing the frozen proposal.

Appendix 9.6 reports the full per-image distribution of these additional oracle gains. This comparison identifies available correction capacity, not the performance of a learned component-wise policy.

5.6. Remaining Frozen-Policy Evaluation

Frozen-policy transfer to fresh shifts. We evaluate two fixed C-lower-bound policies on 200 clean identities disjoint from scorer training and validation. The panel contains an unobserved Gaussian severity, Poisson and speckle noise, spatially varying blur, synthetic snow and raindrops, and both orders of blur–noise composition. No threshold, offset, or candidate is selected on these target outcomes. Table 8

Table 8
Quantitative results on eight fresh synthetic degradation shifts. BaryIR and the rectified-flow proposal are fixed. PSNR (dB) and SSIM are macro-averaged across degradation shifts after averaging within each shift. Parenthesized changes are computed relative to BaryIR before rounding; upward arrows indicate increases. Coverage denotes the fraction of images receiving a nonzero correction, and mean α is the average correction weight. Harmful corrections are executed corrections with $\Delta\mathrm{MSE} > 10^{-12}$. Both τ settings were fixed before evaluation.

Method	PSNR (Δ)	SSIM (Δ)	Coverage (%)	Mean α	Harmful corrections↓
BaryIR	24.09 (0.00)	0.663 (0.000)	–	–	–
Rectified-flow proposal	24.46 (0.37↑)	0.698 (0.035↑)	100.0	1.000	–
GRA-Flow (τ=0.05)	24.29 (0.20↑)	0.670 (0.007↑)	14.9	0.069	0
GRA-Flow (τ=0.10)	24.29 (0.19↑)	0.670 (0.006↑)	14.0	0.066	0

reports absolute restoration quality together with the action statistics required to interpret it. With τ=0.05, the frozen action increases Macro PSNR/SSIM from 24.09/0.663 to 24.29/0.670, while executing on 14.9% of images with mean action 0.069. The τ=0.10 arm yields the same rounded Macro score at a slightly lower coverage and action mass. In this evaluated panel, neither frozen action executes an MSE-increasing correction.

The gain is concentrated where the proposal is useful: on the unseen Gaussian severity, the action changes 17.06/0.295 to 18.48/0.341 (99.0% coverage); it also transfers positively to Poisson noise (24.15/0.620 to 24.32/0.626) and blur–then–noise (22.62/0.514 to 22.64/0.515). It rejects the full proposal on the remaining registered operators. The full proposal's 24.46/0.698 establishes residual correction headroom, whereas the action rows show the attainable target-free portion under one frozen rule. Thus, this panel supports selective refinement on the specified fresh noise and composition shifts; it does not claim uniform improvement over all corruption operators. On paired UIEB (n=90) and LOL-v2-real (n=100) boundary panels, both frozen policies abstain on every image; the action therefore retains the BaryIR Macro score (24.30/0.838) with zero executed harm. This is conservative rejection rather than a new restoration gain. All raw endpoint records and diagnostic controls are retained.

Historical compound-perturbation stress test. We replay the two fixed C candidates on a historical compound-perturbation grid without selecting either candidate on these outcomes. The grid contains 658 paired images; the clean target is unchanged and each already degraded input receives the same SHA-256-determined additive Gaussian perturbation at $\sigma \in \{0, 25, 50, 75, 95\}/255$. Table 9 reports image-weighted restoration scores and paired percentile-bootstrap intervals across images. Both candidates reject every σ=0 case and activate on nearly all cases at $\sigma \geq 50$. At σ=50, τ=0.05 changes the BaryIR endpoint from 17.98/0.350 to 19.24/0.445, a paired gain of 1.264 dB (95% CI [1.189, 1.339]). The gains remain 1.072–1.142 dB at σ=75 and 1.024–1.084 dB at σ=95, whereas the candidates rarely act at σ=25. This result characterizes the response of the frozen policies under this historical stress protocol; it neither selects τ nor replaces a fresh-shift evaluation or a calibrated-risk assessment.

6. Discussion

The central distinction is between generating a correction and knowing whether to apply it. The exact endpoint response makes the required information explicit: proposal energy is observable, while signed alignment with the remaining restoration error must be estimated. A lower score can yield a non-harmful action when its pointwise condition holds, but obtaining that score with useful coverage is a separate learning and calibration problem. The matched normalized-benefit experiments show a pooled sign-ranking advantage for the full trajectory on development identities, but not a corresponding advantage in calibrated correction utility. They also distinguish two sources of rejection: prediction errors that remain when the shared offset is removed, and the additional conservatism induced by the largest environment residuals. The training-versus-OOF comparison further separates fitting error from cross-identity residual tails. Low-light contributes to both in the reported development setting. These observations guide tests of the benefit representation without changing the frozen restoration proposal or adopting dataset-specific thresholds. Paired degradation interventions hold clean identity fixed while changing the observed degradation and measured correction benefit. This supplies a controlled training comparison. Its causal interpretation is limited to those known intervention mechanisms; successful OOD prediction, causal identification, and valid statistical risk certification are not interchangeable conclusions. In particular, a low observed harmful fraction at low correction coverage is not evidence of useful risk control by itself. The final empirical claim requires correction quality, coverage, magnitude, and independent harmful-risk evidence together.

7. Conclusion

We formulate task-free rectified-flow refinement as a baseline-relative continuous correction decision. An exact quadratic response identifies signed correction alignment as the unknown decision quantity, and a valid lower score

Table 9
Historical compound-perturbation stress test on 658 paired images. Gaussian noise with standard deviation $\sigma/255$ is added to each already degraded input while the target remains unchanged. PSNR (dB) and SSIM are image-weighted means. Parenthesized Δ values are paired PSNR differences from BaryIR (dB), with upward/downward arrows indicating increases/decreases. The 95% CI columns report percentile-bootstrap intervals for Δ across images; coverage is the fraction of images receiving a nonzero correction. Both τ candidates were fixed before this evaluation. This is a descriptive historical stress test, not independent OOD certification.

	BaryIR		Ours($\tau = 0.05$)				Ours($\tau = 0.10$)			
σ	PSNR (Δ)	SSIM	PSNR (Δ)	SSIM	95% CI	Coverage	PSNR (Δ)	SSIM	95% CI	Coverage
0	26.05 (0.000)	0.858	26.05 (0.000)	0.858	[0.000, 0.000]	0.0%	26.05 (0.000)	0.858	[0.000, 0.000]	0.0%
25	21.93 (0.000)	0.751	21.92 (0.004↓)	0.751	[-0.008, -0.001]	1.3%	21.92 (0.006↓)	0.751	[-0.011, -0.002]	1.5%
50	17.98 (0.000)	0.350	19.24 (1.264↑)	0.445	[1.189, 1.339]	99.3%	18.96 (0.983↑)	0.421	[0.925, 1.040]	98.4%
75	14.77 (0.000)	0.193	15.85 (1.072↑)	0.232	[1.035, 1.110]	100.0%	15.92 (1.142↑)	0.236	[1.102, 1.183]	100.0%
95	13.05 (0.000)	0.137	14.08 (1.024↑)	0.164	[0.998, 1.051]	100.0%	14.14 (1.084↑)	0.166	[1.056, 1.112]	100.0%

yields an analytic non-harmful action. We implement a target-free trajectory predictor with paired-intervention supervision and examine its location, sign, and lower-tail components through matched development controls. The present evidence localizes prediction and shared-offset bottlenecks while preserving the distinction between deterministic response guarantees and empirical risk certification. The frozen fresh-shift panel establishes selective improvement on specified synthetic shifts; independent risk certification and real-degradation testing remain necessary before claiming risk-controlled OOD gains.

8. Proofs for the Endpoint Correction Rule

Proof of Proposition 1. Expanding the difference of squared errors gives

$$\|r - \alpha d\|_p^2 - \|r\|_p^2 = \alpha^2 \|d\|_p^2 - 2\alpha \langle d, r \rangle_p \tag{15}$$

$$= a\alpha^2 - 2b\alpha. \tag{16}$$

Substitution of $u = \alpha\sqrt{a}$ and $c = b/\sqrt{a}$ gives $u^2 - 2cu$. For $a > 0$, the quadratic is strictly convex and its unconstrained minimizer is b/a; projection onto $[0, 1]$ yields the stated action. Factoring the response as $\alpha(a\alpha - 2b)$ gives the sign conditions. If $a = 0$, then $d = 0$, $b = 0$, and every output equals the baseline.

Proof of Proposition 2. If $\ell \le 0$, then $u = 0$ and the response vanishes. Otherwise $0 < u \le \ell \le c$, so

$$u^2 - 2cu \le u^2 - 2u^2 = -u^2. \tag{17}$$

Thus a harmful accepted correction implies $\ell > c$. For any distribution and fixed predictor/offset, $\Pr(\text{harm}, \alpha > 0) \le \Pr(\ell > c)$. Dividing by the positive acceptance probability proves Eq. (13) under its explicitly assumed marginal bound. No such bound is asserted for the development-reused OOF construction.

Proof of Proposition 3. If $u = 0$, the claim is immediate. Otherwise, $0 < u \le \ell$ and $c \ge \ell - \eta \ge u - \eta$. Substitution into Eq. (5) gives

$$\Delta\text{MSE}(\alpha) = u^2 - 2cu \le -u^2 + 2u\eta. \tag{18}$$

If this upper bound is non-positive, the correction is non-harmful; therefore harm requires $\eta > u/2$. The result is deterministic and does not assert that the learned score obeys this condition on a new distribution.

Table 10
Matched controls on scorer-training OOF predictions. PSNR reports absolute Macro PSNR (dB), computed from per-image MSE before environment averaging; differences are calculated before rounding. Parenthesized Δ values are equal-environment mean per-image PSNR changes (dB) relative to the frozen baseline; upward/downward arrows indicate increases/decreases; Coverage denotes (%); Mean action is the all-image mean of α; Harm is the pooled harmful fraction among accepted corrections (%). A dash denotes an inapplicable baseline selector statistic or an undefined harmful fraction because no correction is accepted. The mean-only control does not use τ; its repeated entries denote the same result. These are development diagnostics, not held-out OOD results.

	$\tau = 0.05$				$\tau = 0.10$			
Control	PSNR (Δ)	Coverage (%)	Mean α	Harm (%)	PSNR (Δ)	Coverage (%)	Mean α	Harm (%)
Frozen baseline	25.32 (0.000)	–	–	–	25.32 (0.000)	–	–	–
Full C predictor	25.40 (0.077↑)	4.35	0.0152	3.45	25.38 (0.055↑)	3.85	0.0115	5.19
w/o paired-intervention loss	25.42 (0.094↑)	4.85	0.0194	4.12	25.37 (0.050↑)	3.70	0.0110	4.05
w/o sign loss	25.40 (0.080↑)	4.35	0.0165	4.60	25.37 (0.041↑)	3.75	0.0094	5.33
w/o pinball loss	25.32 (0.000)	0.00	0.0000	–	25.32 (0.000)	0.00	0.0000	–
Mean-only C predictor	25.32 (0.001↓)	0.30	0.0025	66.67	25.32 (0.001↓)	0.30	0.0025	66.67
Endpoint-only C predictor	25.45 (0.128↑)	4.80	0.0223	0.00	25.39 (0.070↑)	3.85	0.0120	0.00

9. Additional Calibration and Fitting Diagnostics

9.1. Trajectory Observables and Residual-Offset Details

The frozen ten-step sampler evaluates its reference and increment fields and produces a clipped proposal endpoint. At each step, we collect nine scalar summaries: state RMS; reference, increment, and combined-velocity RMS; reference–increment inner product and cosine; update RMS; and velocity-change RMS and consecutive-velocity cosine. The two consecutive-velocity statistics are zero at the first step. Input–baseline RMS and endpoint-correction RMS complete the 92-dimensional vector. All summaries are computed from the same proposal pass without the clean target or an environment label. For the 400 OOF records per environment, the order statistic in Eq. (10) is $\lceil 401 \times 0.95 \rceil = 381$. Every held-fold prediction is transformed back with its training-fold target statistics before residuals are combined. The shared offset is computed in these original normalized-benefit units. These lowercase image-level quantities are distinct from the environment-level velocity moments used in the historical shared-guidance study.

9.2. Complete Auxiliary-Loss Controls

9.3. Mechanism Diagnosis of Baseline-Relative Selective Correction

We examine how benefit prediction and a shared calibration offset determine whether a frozen restoration proposal is applied. This development study concerns image-space endpoint corrections, rather than environment-level velocity moments. We retain all 400 clean-image identities and their five paired degradation environments (2,000 records per candidate), using five identity-grouped folds for out-of-fold (OOF) prediction. The proposal and 92 target-free input features are fixed. Both quantile settings, $\tau \in \{0.05, 0.10\}$, are reported; no held-out OOD outcomes enter this diagnosis. For a baseline output x_i^0, a frozen correction d_i, and target y_i, define $a_i = \|d_i\|^2$, $b_i = \langle d_i, y_i - x_i^0 \rangle$, and $c_i = b_i/\sqrt{a_i}$, with pixel-averaged inner products. The change in MSE at correction weight α_i is $a_i\alpha_i^2 - 2b_i\alpha_i$. Given a fitted lower-tail score q_i for c_i, we use $\alpha_i = \mathrm{clip}((q_i - \delta)/\sqrt{a_i}, 0, 1)$ and reject zero-energy proposals. The shared offset δ is the maximum, across the five environments, of the 381st ordered residual $q_i - c_i$ among 400 OOF records. These are the per-image quantities defined in Eq. (4). Table 11 separates rejection induced by the shared offset from errors already present in the predicted correction score. At $\tau = 0.05$, removing the offset changes blur coverage from 0.00% to 42.75%, with six harmful corrections among 171 accepted proposals. The same intervention accepts only 12 JPEG proposals, of which ten are harmful. Thus, reducing rejection alone does not resolve benefit discrimination across degradations. The second quantile setting reproduces this distinction: blur coverage reaches 55.00%, whereas 18 of the 27 accepted JPEG corrections increase MSE. Low-light residuals determine the shared offset at both settings (0.040831 and 0.048317). At $\tau = 0.05$, the low-light location-prediction absolute error is 0.031178, compared with 0.009249 for Gaussian noise. Among the 20 largest low-light residuals, the average location-prediction error is 0.102378, exceeding the average predicted location-to-tail gap of 0.030714. Eighteen of these identities recur among the corresponding 20 records at $\tau = 0.10$. These observations motivate a targeted examination of benefit prediction and tail calibration while retaining the same restoration proposal. Because offsets and diagnostics use the same pooled OOF outcomes, the reported rates are descriptive development results, not held-out risk certificates or evidence of OOD generalization.

Table 11
OOF mechanism diagnosis with frozen predictions. Coverage is the percentage of records with $\alpha_i > 0$. Zero-offset harm is the percentage of accepted corrections that increase MSE by more than 10^{-12}; parentheses give harmful/accepted counts. Setting $\delta = 0$ is a diagnostic intervention, not a proposed deployment policy. Each environment contains 400 paired identities.

	$\tau = 0.05$			$\tau = 0.10$		
Environment	Shared coverage	Zero-offset coverage	Zero-offset harm	Shared coverage	Zero-offset coverage	Zero-offset harm
Gaussian blur	0.00%	42.75%	3.51% (6/171)	0.00%	55.00%	3.18% (7/220)
Gaussian noise	21.25%	44.75%	7.26% (13/179)	18.50%	49.75%	9.05% (18/199)
JPEG	0.00%	3.00%	83.33% (10/12)	0.00%	6.75%	66.67% (18/27)
Low-light gamma	0.25%	3.00%	25.00% (3/12)	0.00%	7.75%	35.48% (11/31)
Spatial haze	0.25%	5.25%	76.19% (16/21)	0.75%	8.50%	76.47% (26/34)

Table 12
Environment-level mean-only diagnosis on OOF development predictions. Each environment contains 400 images. Absolute PSNR is averaged over all images, including rejected proposals; changes are computed before rounding. Absolute PSNR is reported in dB; parenthesized Δ values are PSNR changes in units of 10^{-3} dB, with upward/downward arrows indicating increases/decreases. Accepted counts are out of all images, and harmful counts are out of accepted corrections. The last column gives the environment-specific residual offset before taking the shared maximum; these offsets are diagnostics, not environment-conditioned deployment rules.

Environment	Baseline PSNR (dB)	Mean-only PSNR (Δ)	Accepted / total	Harmful / accepted	Residual offset
Gaussian blur	25.42	25.42 (0.000)	0/400	–	0.021
Gaussian noise	27.83	27.83 (0.095↑)	3/400	1/3	0.016
JPEG	26.91	26.91 (0.000)	0/400	–	0.026
Low-light (gamma)	17.50	17.50 (3.049↓)	2/400	2/2	0.079
Spatial haze	28.95	28.95 (0.724↓)	1/400	1/1	0.045

9.4. Environment-Level Mean-Only Control

Mean-only estimation does not resolve shared-offset rejection. We additionally retain only mean-regression supervision, disable the sign, pairwise, and pinball losses, and use the predicted mean directly in the residual-offset construction. This avoids interpreting an unsupervised quantile output, while retaining the frozen proposal, features, folds, and training budget. Unlike the single-loss ablations, this is a jointly modified mean-based control. The quantile parameter is inactive, so its identical entries in Table 10 are not independent repetitions.

The residual tail remains the discriminating quantity. Table 12 localizes the outcome: low-light supplies the largest residual offset (0.07902), which is then shared across environments. Only 6 of 2,000 corrections are accepted, with an all-image correction mass of 0.0025. Four of these six corrections are harmful; the resulting 66.67% is a descriptive fraction from six accepted cases, not a population risk estimate. Gaussian noise contributes a small positive PSNR change, whereas low-light and haze account for the negative Macro change. Thus, substituting a mean-based residual estimate does not by itself recover useful correction coverage in this setting. The fitting-versus-OOF analysis below examines this distinction without changing the acceptance threshold. These development results do not establish held-out OOD performance or isolate any one removed loss.

9.5. Fitting Error versus Cross-Identity Residual Tails

We replay the saved fold checkpoints to separate training fit from excluded-identity prediction. The completed diagnostic covers 50 checkpoints and reproduces the saved OOF predictions with zero maximum replay discrepancy; it opens no held-out partition. Figure 5 shows the full predictor at both fixed quantiles. At $\tau = 0.05$, low-light location-prediction MAE increases from 0.020906 on training identities to 0.031178 on excluded identities; the mean fold-level 95th residual percentile increases from 0.010310 to 0.034903. The analogous residual values at $\tau = 0.10$ are 0.024967 and 0.048375. The training errors and the additional excluded-identity tail both contribute to the diagnostic picture. These observations localize the next representation comparison to directional-benefit prediction and its upper residual tail; they do not justify changing the shared acceptance rule or opening independent tests for model selection.

9.6. Distribution of Additional Oracle Capacity

Figure 6 distinguishes a broader improvement from a mean driven by a few images. Separate amplitudes exceed the shared oracle by more than 0.05 dB on 149/400 haze images (37.25%), compared with 40/400 low-light images (10.00%). The corresponding mean/median extra gains are 0.0967/0.0142 dB and 0.0172/0.0003 dB. The benefit is therefore heterogeneous and more widespread in haze. Because the shared action is nested within the separate action

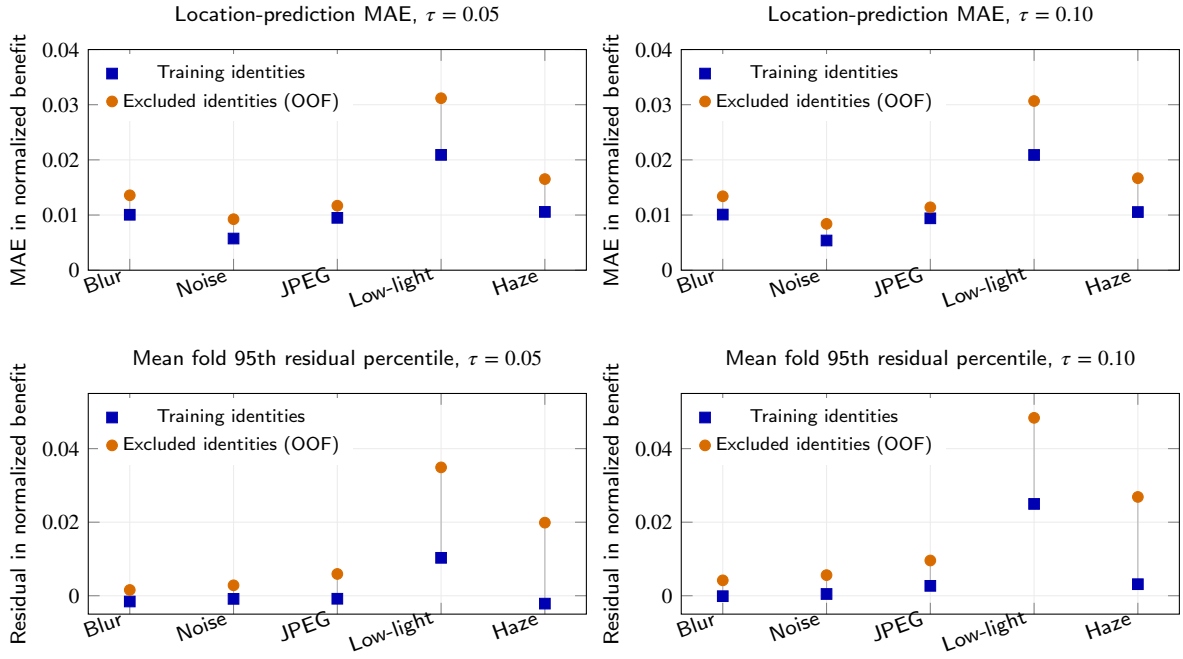


Fig. 5. Development diagnosis of fitting and cross-identity error for the full normalized-benefit predictor. Each connector pairs training and excluded-identity statistics for the same environment, not different degradation categories. Each point averages five fold-level statistics; the lower panels average fold-specific 95th percentiles rather than taking a pooled percentile. The 400 clean identities yield 2,000 degraded records per candidate. Training identities occur in multiple fitted models; points are descriptive, not independent repetitions or confidence intervals. Both fixed quantile settings and all five environments are shown.

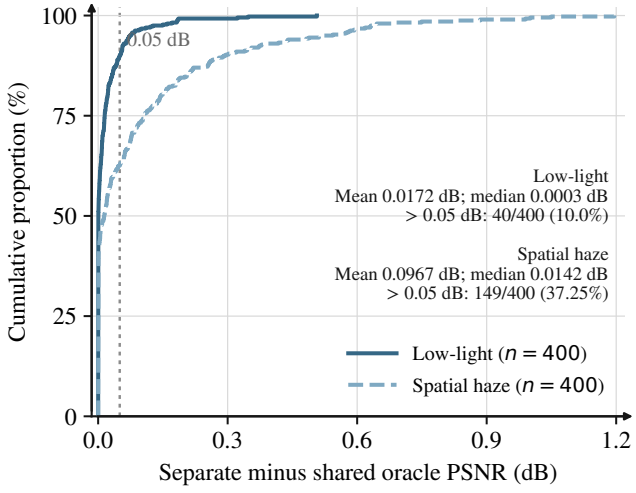


Fig. 6. Distribution of per-image PSNR gains from separate versus shared oracle amplitudes. Each empirical cumulative distribution retains all 400 images, including zero gains, on identical axes. A lower curve at a given gain indicates more images exceeding that gain; dotted lines mark 0.05 dB. Both oracles use clean targets, and the separate action set contains the shared one. The comparison measures the size and distribution of additional oracle capacity, not learned prediction performance.

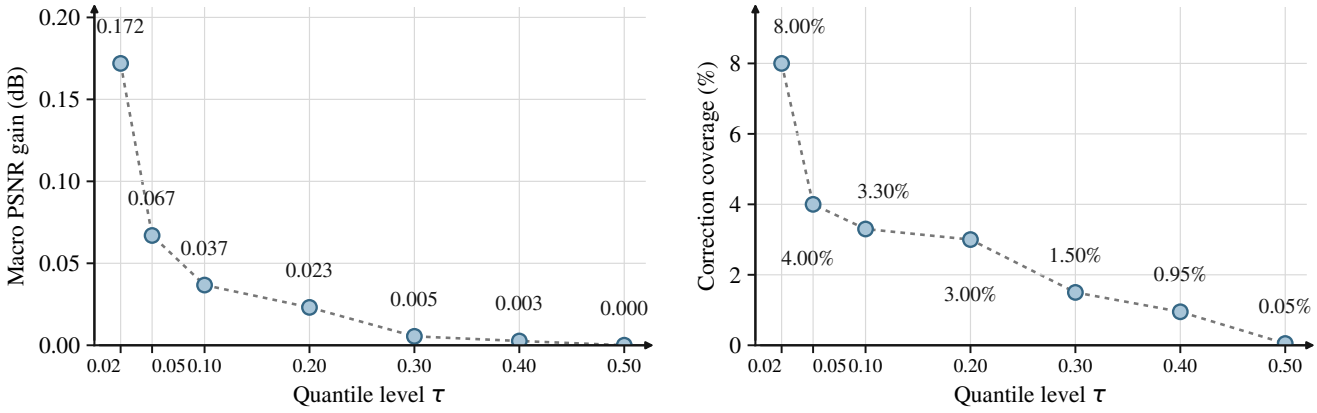
set and both allow zero correction, non-increasing oracle MSE follows by construction. The empirical evidence is the magnitude and prevalence of the additional gain, rather than the inequality itself. This diagnosis motivates testing whether target-free component-benefit estimates can recover that capacity; it does not establish causal factor identification or independent OOD generalization.

10. Additional Development Diagnostics

We retain the same frozen baseline and A+B proposal throughout these additional controls. The observations below concern the 400 scorer-training identities, not an independent OOD test. All-image PSNR includes rejected images at their baseline output, and all five environments are retained.

10.1. Quantile Parameterization and Residual Scaling

The direct-quantile control predicts an unconstrained scalar instead of subtracting a nonnegative gap from the location output, while preserving the location, sign, paired, and pinball training losses. Figure 7 reports every evaluated direct-head quantile, $\tau \in \{0.02, 0.05, 0.10, 0.20, 0.30, 0.40, 0.50\}$. Lowering τ to 0.02 yields 25.4968 dB Macro PSNR (+0.1719 dB), 160 accepted cases, and zero observed harmful cases; coverage remains 8.00%. This is the largest gain among the reported direct-head settings, not an independently selected operating point. At the same $\tau = 0.02$, the coupled-head control yields 25.4481 dB (+0.1232 dB), with 3 harmful cases among 117



(a) Direct-quantile utility

(b) Direct-quantile coverage

Fig. 7. Complete direct-quantile sensitivity on scorer-training OOF records. Each point is a separately fitted quantile setting using the same proposal, features, folds, losses, and training budget, followed by its shared development residual offset. All seven evaluated settings are shown. Lines connect the ordered quantile settings without smoothing; they do not imply a continuous optimum or an independent risk guarantee.

Table 13
Residual-scale control with fixed direct-quantile predictions at $\tau = 0.02$. Both controls use all 2,000 OOF development records; PSNR averages all images. Harmful/accepted counts are shown explicitly.

Statistic	Constant	Learned
PSNR (dB)	25.50	25.61
ΔP (dB)	+0.1719	+0.2845
Cov. (%)	8.00	9.30
Mass	0.0334	0.0743
Harmful / accepted	0/160	6/186

accepted corrections. The paired intervals for sign AUPRC, location MAE, and location-derived action regret include zero in this head comparison; a difference in selected correction output is not evidence that one parameterization has superior benefit representations.

We also keep the direct $\tau = 0.02$ predictions fixed and learn a positive residual scale from nested identity-held-out training residuals. The resulting adjustment has the form $q - \delta s(X)$ rather than $q - \delta$. Table 13 reports the all-image change alongside raw accepted and harmful counts. The gain increases from +0.1719 to +0.2845 dB, with the largest contribution coming from Gaussian noise (+1.4248 dB under learned scaling). The same learned-scale policy gives changes of +0.0061, -0.0085, -0.0001, and 0.0000 dB for blur, JPEG, low-light, and haze, respectively. Low-light continues to determine the shared standardized residual offset. The result demonstrates an effect of residual scaling on the operating point, not resolution of benefit discrimination across all environments.

10.2. Dedicated Classification and Spatial Observables

The dedicated sign control in Table 3 uses BCE alone with the same hidden widths and training budget as the auxiliary predictor. Pooled sign AUPRC is 0.8487 for dedicated training and 0.8560 for the existing auxiliary head. The complete threshold curves in Fig. 1 therefore accompany fixed-cutoff results; different coverage at the same numerical score cutoff is not interpreted as a capacity-matched improvement in the full correction system.

To test the information lost by global aggregation, we construct 18 target-free endpoint maps: signed RGB correction, squared correction, correction times the input–baseline residual, a 5×5 low-pass correction, its high-pass complement, and correction-gradient times baseline-gradient alignment. Spatial averaging over a nonoverlapping 4×4 grid gives 288 additional coordinates. These describe endpoint directions, not unobserved local trajectories or identified causal variables. The three controls concatenate the original 92 features with zeros, the same map means broadcast across 16 cells, or the local grid values. All have a 380-dimensional interface and identical allocated network architecture, while their effective input information differs. We train each with sign BCE and keep the action amplitudes fixed.

Table 14 reports pooled and per-environment AUPRC. Local descriptors raise low-light AUPRC from 0.5579 to 0.6505. At $p_0 = 0.5$, both controls accept 180 low-light images, and harmful counts change from 86 to 70. JPEG AUPRC changes from 0.4808 to 0.2976, so this low-light observation does not establish an overall spatial-representation advantage. Table 15 retains both cutoffs and all-image outcomes. The original-feature control here is padded and retrained under the matched 380-input architecture; it is not the earlier 92-input dedicated classifier. Before feature extraction, the original baseline and proposal output hashes, 92 features, response labels, and metrics were reproduced

Table 14
Endpoint-spatial feature diagnostic: sign AUPRC on fixed OOF development records. Each environment contains 400 images. The pooled row combines all 2,000 records and is not an average of the environment-specific AUPRC values. Positive fractions are reported as class-balance context.

Environment	Positive fraction	Original + zero padding	Additional global means	Local direction grid
Blur	0.9825	0.9911	0.9870	0.9744
Noise	0.8100	0.9686	0.9751	0.9728
JPEG	0.1675	0.4808	0.4700	0.2976
Low-light	0.4000	0.5579	0.5497	0.6505
Haze	0.4500	0.4886	0.5095	0.4919
Pooled	0.5620	0.8552	0.8411	0.8239

Table 15
Complete fixed-cutoff outcomes for the spatial feature diagnostic. All controls use the same 2,000 development images and frozen action amplitudes. PSNR and ΔP are absolute and baseline-relative Macro scores in dB; Cov. is all-image coverage and Mass is mean α. Scores are not assumed to be calibrated probabilities.

$p_0 = 0.5$			
Statistic	Original + zero padding	Additional global means	Local direction grid
PSNR	25.66	25.66	25.66
ΔP	+0.3338	+0.3327	+0.3353
Cov. (%)	55.40	56.75	55.90
Mass	0.3048	0.2984	0.2908
Harmful / accepted	281/1108	266/1135	279/1118
$p_0 = 0.9$			
Statistic	Original + zero padding	Additional global means	Local direction grid
PSNR	25.66	25.66	25.66
ΔP	+0.3369	+0.3368	+0.3368
Cov. (%)	27.85	45.15	49.65
Mass	0.2219	0.2713	0.2742
Harmful / accepted	52/557	175/903	227/993

exactly for all 2,000 records using the original inference runtime.

11. Supplementary Historical Restoration Results

We retain the complete historical comparison tables to preserve the numerical record. The shared-guidance variants and MIR-Flow predecessor must not be interpreted as the current Gain-Risk Aligned Flow policy. The stress-test grid has already informed development. The legacy five-degradation rows require Run-level attribution before they can support a current-method comparison; no numeric entry is replaced during this reorganization. Historical scale sensitivity concerns w, not the endpoint action α.

Qualitative comparison. Figure 8 complements the historical restoration results with matched visual comparisons. The denoising example uses the preselected Kodak24 image `kodim03` with $\sigma = 25/255$ and a common 496×496 crop; every method receives the same saved noisy input. The other four examples retain the original images and outputs. In this figure, BaryIR and Ours denote the original backbone and predecessor-refinement checkpoints, respectively.

Table 16
Historical compound-OOD development stress test. Gaussian noise with standard deviation $\sigma \in \{0, 25, 50, 75, 95\}/255$ is added to each already degraded OOD input, while the target remains unchanged. All methods use the same 658 paired images, deterministic per-image perturbations, and RGB scoring. The GRA column is the historical shared-guidance variant, not the current endpoint-selection policy. These previously inspected data are not untouched confirmation evidence.

Dataset	σ	PromptIR		DA-CLIP		DiffUIR		AdaIR		MoCE-IR		DA-RCOT		BaryIR		B1 control		GRA (legacy)	
		PSNR (dB)	SSIM	PSNR (dB)	SSIM	PSNR (dB)	SSIM	PSNR (dB)	SSIM	PSNR (dB)	SSIM	PSNR (dB)	SSIM	PSNR (dB)	SSIM	PSNR (dB)	SSIM	PSNR (dB)	SSIM
SOTS-Outdoor	0	29.95	0.977	30.58	0.956	**32.00**	**0.980**	29.77	0.977	30.10	0.975	30.36	0.977	29.55	0.974	29.44	0.970	29.45	0.970
	25	15.34	0.703	15.25	0.627	13.66	0.446	15.36	0.705	**15.37**	0.707	15.37	0.706	15.36	0.707	15.03	0.710	15.04	**0.712**
	50	15.18	0.640	**15.28**	0.586	12.36	0.452	15.22	0.642	15.24	**0.645**	12.38	0.136	14.31	0.279	13.87	0.241	14.07	0.264
	75	14.83	**0.529**	12.79	0.137	12.24	0.405	**14.87**	0.332	14.76	0.381	10.36	0.075	12.67	0.142	12.26	0.127	12.55	0.138
	95	**14.33**	**0.364**	11.52	0.092	12.12	0.343	13.52	0.161	13.77	0.191	9.39	0.055	11.55	0.097	11.17	0.088	11.47	0.096
JPEG	0	27.57	**0.825**	27.09	0.798	27.36	0.823	**27.88**	0.825	26.78	0.818	27.02	0.819	26.61	0.815	26.32	0.801	26.33	0.802
	25	28.53	**0.819**	26.29	0.730	15.71	0.583	**28.53**	0.818	28.51	0.818	28.47	0.817	28.51	0.818	28.18	0.813	28.20	0.814
	50	27.26	0.767	25.72	0.712	15.21	0.498	**27.28**	**0.768**	27.25	0.767	15.94	0.207	20.95	0.383	19.94	0.342	20.63	0.369
	75	**24.57**	**0.625**	16.35	0.207	15.34	0.400	22.56	0.455	23.02	0.490	12.37	0.117	16.32	0.211	15.60	0.192	16.17	0.207
	95	**21.69**	**0.454**	14.07	0.142	15.22	0.314	18.10	0.247	19.04	0.287	10.94	0.088	14.11	0.149	13.53	0.137	14.01	0.146
Raindrop	0	22.98	0.827	**30.00**	**0.890**	23.49	0.836	23.08	0.827	22.36	0.814	22.13	0.812	23.09	0.829	23.00	0.818	23.04	0.823
	25	23.40	0.748	23.06	0.666	14.69	0.520	23.41	0.749	23.41	0.751	23.40	0.750	**23.41**	0.750	23.39	0.754	23.40	**0.755**
	50	23.05	0.693	22.71	0.654	15.11	0.478	**23.05**	0.694	23.05	**0.695**	15.13	0.137	19.33	0.287	18.54	0.251	19.09	0.276
	75	**21.79**	**0.554**	15.77	0.140	15.35	0.408	20.65	0.362	20.84	0.391	11.97	0.075	15.54	0.143	14.90	0.129	15.41	0.140
	95	**19.94**	**0.370**	13.68	0.094	14.97	0.325	17.18	0.177	17.88	0.206	10.62	0.055	13.57	0.097	13.04	0.089	13.48	0.096
Snow	0	22.80	0.795	**30.56**	0.902	30.28	**0.915**	23.01	0.794	22.82	0.784	22.66	0.773	22.85	0.793	22.66	0.784	22.67	0.784
	25	21.47	0.728	20.58	0.648	19.29	0.501	21.47	0.728	21.45	0.727	21.46	0.728	**21.47**	**0.728**	21.29	0.723	21.31	0.724
	50	20.96	**0.682**	20.19	0.602	13.66	0.416	**20.98**	0.682	20.94	0.679	14.87	0.243	18.30	0.407	17.69	0.373	18.09	0.396
	75	**19.86**	**0.576**	15.28	0.240	13.75	0.365	19.13	0.439	19.19	0.475	11.83	0.143	15.12	0.242	14.55	0.223	14.99	0.238
	95	**18.41**	**0.443**	13.38	0.171	13.59	0.308	16.39	0.266	16.84	0.302	10.54	0.108	13.34	0.176	12.85	0.162	13.25	0.173
Macro	0	25.82	0.856	**29.56**	0.886	28.28	**0.889**	25.93	0.856	25.51	0.848	25.54	0.845	25.53	0.853	25.35	0.843	25.37	0.845
	25	22.18	0.749	21.29	0.668	15.84	0.512	**22.19**	0.750	22.18	0.751	22.17	0.750	22.19	0.751	21.97	0.750	21.99	**0.751**
	50	21.61	0.696	20.97	0.638	14.08	0.461	**21.64**	0.696	21.62	**0.697**	14.58	0.181	18.22	0.339	17.51	0.302	17.97	0.326
	75	**20.26**	**0.571**	15.05	0.181	14.17	0.395	19.30	0.397	19.45	0.434	11.63	0.102	14.91	0.185	14.33	0.168	14.78	0.181
	95	**18.59**	**0.408**	13.16	0.125	13.98	0.322	16.30	0.213	16.88	0.247	10.37	0.076	13.14	0.130	12.65	0.119	13.05	0.128

Table 17
Historical five-degradation all-in-one comparison in the BaryIR reporting format. The last row reproduces the predecessor numbers also shown in Table **??**; these are not results of the current endpoint-selection policy. Exact Run attribution must be reconciled before this mixed-source table supports a comparative claim. All numeric entries are retained; ranking marks are omitted pending that reconciliation.

Method	Venue	*Dehazing* SOTS		*Deraining* Rain100L		*Denoising* BSD68$\sigma = 25$		*Deblurring* GoPro		*Low-light* LOL-v1		Average	
		PSNR (dB)	SSIM	PSNR (dB)	SSIM	PSNR (dB)	SSIM	PSNR (dB)	SSIM	PSNR (dB)	SSIM	PSNR (dB)	SSIM
Restormer	CVPR'22	24.09	0.927	34.81	0.971	30.78	0.876	27.22	0.829	20.41	0.806	27.46	0.881
PromptIR	NeurIPS'23	30.41	0.972	36.17	0.970	31.20	0.885	27.93	0.851	22.89	0.829	29.72	0.901
DA-CLIP	ICLR'24	29.78	0.968	35.65	0.962	30.93	0.885	27.31	0.838	21.66	0.828	29.07	0.896
DiffUIR	CVPR'24	29.47	0.965	35.98	0.968	31.02	0.885	27.50	0.845	22.32	0.826	29.25	0.898
InstructIR	ECCV'24	27.10	0.956	36.84	0.973	31.40	0.890	29.40	0.886	23.00	0.836	29.55	0.908
AdaIR	ICLR'25	30.54	0.978	38.02	0.981	31.35	0.889	28.12	0.858	23.00	0.845	30.20	0.910
MoCE-IR	CVPR'25	30.48	0.974	38.04	0.982	31.34	0.887	30.05	0.899	23.00	0.852	30.25	0.919
DA-RCOT	TPAMI'25	30.96	0.975	37.87	0.980	31.23	0.888	28.68	0.872	23.25	0.836	30.40	0.911
BaryIR$_{\text{Restormer}}$	TPAMI'26	31.20	0.979	38.10	0.982	31.43	0.891	29.51	0.889	23.37	0.854	30.72	0.919
BaryIR$_{\text{PromptIR}}$	TPAMI'26	31.68	0.980	38.36	0.984	31.49	0.894	29.84	0.895	23.88	0.862	31.05	0.923
Frozen backbone	Ours	30.13	0.977	38.01	0.981	31.72	0.898	32.24	0.929	23.44	0.851	31.11	0.927
Legacy refinement	Ours	30.81	0.976	38.66	0.983	31.77	0.900	32.36	0.932	23.51	0.854	31.42	0.929

Table 18
Historical shared-guidance-scale robustness under the fixed 11-point grid $w \in \{0, 0.5, \ldots, 5.0\}$. Normalized AUC is computed from Macro PSNR; lower worst-scale drop indicates less sensitivity.

Method	Normalized AUC	Worst-scale drop
B1 matched control	25.0692	0.9349
GRA (legacy)	**25.3108**	**0.0420**

Table 19
Legacy MIR-Flow results on four unseen degradation types. Each entry is PSNR/SSIM, and deltas were computed before display rounding. Historical SSIM uses the predecessor evaluator and is not compared with formal GRA-Flow SSIM. Sample counts are 500 (SOTS-Outdoor), 276 (JPEG), 58 (Raindrop), and 2,000 (Snow); the Average has no separate sample count.

Dataset	Backbone	MIR-Flow	Δ
SOTS-Outdoor	30.35/0.976	31.02/0.975	+0.67/−0.0011
JPEG	26.01/0.813	26.24/0.813	+0.23/+0.0004
Raindrop	23.12/0.830	23.22/0.829	+0.11/−0.0009
Snow	22.89/0.761	22.96/0.762	+0.08/+0.0008
Average	25.59/0.845	25.86/0.845	+0.27/−0.0003



Fig. 8. Qualitative comparison for dehazing, deraining, denoising, deblurring, and low-light enhancement (top to bottom). Within each row, all methods use the same image, display crop, highlighted region, and magnification. Ours denotes the predecessor refinement used in this historical comparison.

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